lecture ②:

Determination of the Motion of a Particle

$\int 2x\,dx$ ← w.r.t t

Symbol integral — with respect to t

$v = \frac{ds}{dt}$ ① $a = \frac{dv}{dt}$ @ $\frac{ds}{ds}$ $a = v\frac{dv}{ds}$ ②

Req → (s, t)
Given → $v = v(t)$
$v = v(s)$
$v = \text{const}$

Req → (v, t)
Given → $a = a(t)$
$a = a(v)$
$a = a_c$ const

Req → (v, s)
Given → $a = a(s)$
$a = a(v)$
$a = a_c$ const

Acceleration Types

Const $= a_c$

$v = v_0 + a_c t$

$v^2 = v_0^2 + 2a_c(s - s_0)$

$s - s_0 = v_0 t + \frac{1}{2} a_c t^2$

Variable
$a(t), a(v), a(s)$

Solution
① Rule Determine
② substi
③ sep
④ Integral

$a = a(t)$ use Rule ② → $a = \frac{dv}{dt}$ → $a(t) = \frac{dv}{dt}$

$\int dv = \int a(t)\,dt$

$v = \int a(t)\,dt$

(2) $a = a(s)$ use Rule (3) $\rightarrow a = v\frac{dv}{ds}$ بترتيب $\rightarrow a(s) = v\frac{dv}{ds}$

بالتكامل $\rightarrow \int v\,dv = \int a(s)\,ds \;\rightarrow\; \frac{v^2}{2} = \int a(s)\,ds$

(3) $a = a(v)$ ~~use Rule (2)~~ $\rightarrow a = \frac{dv}{dt} \rightarrow a(v) = \frac{dv}{dt}$

بداية عند السكون ∴ $S_0 = 0m$

Example ①

A bicyclist starts from rest and after traveling along a straight path a distance of 20 m reaches a speed of 36 km/h. Determine his acceleration if it is constant. Also, the time of travel.

$$a = a_c = Const$$

→ starts from rest

$S_0 = 0m$

$v_0 = 0 m/s$

$t_0 = 0s$

→ $S = 20m$ أي قطع

وصل لسرعة $V = 36 km/h$ *نتذكر $\frac{5}{18}$ للتحويل لـ m/s

$\therefore V = 36 * \frac{5}{18} = 10 m/s$

→ ① مطلوب أيجاد a_c

② مجهول زمن الرحلة t

$V^2 = V_0^2 + 2a_c(S - S_0)$

~~$36^2 = 0 + 2a_c(20 - 0)$~~

~~$1296 = 2 * 20 a_c$~~

~~$1296 = 40 a_c$~~

$10^2 = 0^2 + 2a_c(20 - 0)$

$100 = 40 a_c$

$a_c = 2.5 m/s^2$

$V = V_0 + a_c t$

$10 = 0 + 2.5 * t$

$t = 4$

هنا معايا

S_0, V_0, S, V, t_0

وعايز a_c و t

بسيب استخدم معادلة السرعة

$V^2 = V_0^2 + 2a_c(S - S_0)$

بعد ما جبت a_c استخدم

معادلة السرعة $V = V_0 + a_c t$

لإيجاد الزمن

Example (2)

The acceleration of a particle along a straight line is defined by $a = 2t - 9\ m/s^2$, where t is in seconds. At $t=0$, $s = 1\ m$ and $v = 10\ m/s$. When $t = 9$, determine (a) the particle's position, (b) the total distance traveled, and (c) the velocity.

a is variable in t $\boxed{a = a(t)}$

$t_0 = 0 \Rightarrow s_0 = 1\ m,\ v_0 = 10\ m/s$

$t = 9 \Rightarrow s\ ,\ D_t\ ,\ v$

$a = \frac{dv}{dt} \longrightarrow 2t - 9 = \frac{dv}{dt}$

$\int dv = \int 2t - 9\ dt \Rightarrow \boxed{v = t^2 - 9t + C_1}$

$v = t^2 - 9t + C_1$, $t = 0 \Rightarrow 10 = 0 - 0 + C_1 \Rightarrow C_1 = 10$

$\therefore v = t^2 - 9t + 10$

$t = 9 \Rightarrow v(9) = 9^2 - 9*9 + 10 \Rightarrow v(9) = 10\ m/s$

$v = t^2 - 9t + 10 \longrightarrow v = \frac{ds}{dt} \longrightarrow t^2 - 9t + 10 = \frac{ds}{dt}$

$\int ds = \int t^2 - 9t + 10\ dt \Rightarrow s = \frac{t^3}{3} - \frac{9}{2}t^2 + 10t + C_2$

$t = 0 \Rightarrow 1 = 0 - 0 + 0 + C_2 \Rightarrow C_2 = 1$

$\therefore s_{(t)} = \frac{t^3}{3} - \frac{9}{2}t^2 + 10t + 1$

$t = 9 \Rightarrow s_{(t)} = \frac{9^3}{3} - \frac{9}{2}*9^2 + 10*9 + 1 = -30.5\ m$

$v = t^2 - 9t + 10 = 0 \Rightarrow t \simeq 7.7\ ,\ t \simeq 1.29$

$s_{(0)} = 1\ m$, $s_{(1.29)} = \frac{(1.29)^3}{3} - \frac{9}{2}*(1.29)^2 + 10*1.29 + 1 \simeq 7.12$

$s_{(7.7)} = \frac{(7.7)^3}{3} - \frac{9}{2}*(7.7)^2 + 10*7.7 + 1 \simeq -36.12\ m$

$s_{(9)} = -30.5\ m$

$|s_{1.29} - s_0| \simeq 6.12$, $|s_{7.7} - s_{1.29}| \simeq 43.24$, $|s_9 - s_{7.7}| \simeq 5.62$

$D_t \simeq 55\ m$

Example ③

A particle travels in a straight line with accelerated motion such that $a = -ks$, where s is the position and k is a proportionality constant which is to be determined. For $s_0 = 2\,m$, the velocity is $4\,m/s$, and for $s = 3.5\,m$ the velocity is $10\,m/s$. What the position s when the velocity $v = 0$?

$a = a(s) = -ks$

$a = v\frac{dv}{ds}$

$a = v\frac{dv}{ds} \xrightarrow{\text{تعويض}} -ks = v\frac{dv}{ds} \xrightarrow{\text{فصل المتغيرات}} \int v\,dv = \int -ks\,ds$

$\xrightarrow{\text{التكامل}} \frac{v^2}{2} = -k\frac{s^2}{2} + C$

Given

for $s_0 = 2\,m \longrightarrow v = 4\,m/s$, for $s = 3.5\,m \longrightarrow 10\,m/s$

$\frac{4^2}{2} = -k\frac{2^2}{2} + C \quad , \frac{10^2}{2} = -k\frac{3.5^2}{2} + C$

$8 = -2k + C \quad , 50 = -6.125k + C$

$\therefore k = -10.18 \quad , C = -12.36$

what the position when the velocity $v = 0$?

$\frac{v^2}{2} = -k\frac{s^2}{2} + C \Rightarrow 0 = \frac{+10.18}{2}s^2 - 12.36$

$5.09\,s^2 - 12.36 = 0$

$5.09\,s^2 = 12.36$

$s^2 = 2.428$

$s = \pm 1.558\,m \Rightarrow s = 1.558\,m \;,\; s = -1.558\,m$

Scanned with CamScanner

Example (4.)

A small projectile fired vertically downward into a fluid medium with an initial velocity of 60 m/s. Due to the resistance of the fluid, the projectile experiences a deceleration equal to $a = -0.4v^3 \ m/s^2$, where v is in m/s. Determine the projectile's velocity and position 4s after it is fired.

يتم إطلاق مذنبة صغيرة رأسيًا إلى أسفل داخل وسط مائع بسرعة ابتدائية مقدارها 60 m/s ومتعرضة لمقاومة المائع، تتعرض المذنبة لتباطؤ يساوي $a = -0.4v^3 \ m/s^2$ حيث v مقاسة بـ m/s. احسب سرعة المذنبة وموقعها بعد 4 ثوانٍ من إطلاقها.

$$a = -0.4v^3$$

$$a = \frac{dv}{dt} \xrightarrow{\text{بالتعويض}} -0.4v^3 = \frac{dv}{dt} \xrightarrow{\text{فصل المتغيرات}} -0.4v^3 dt = dv \Rightarrow \frac{1}{v^3} dv = -0.4\, dt$$

$$\xrightarrow{\text{بأخذ التكامل}} \int_{60}^{v} v^{-3}\, dv = \int_{0}^{t} -0.4\, dt \Rightarrow \frac{v^{-2}}{-2}\Big|_{v_0}^{v} = -0.4t\Big|_{t_0}^{t}$$

$$\Rightarrow -\frac{1}{2v^2}\Big|_{60}^{v} = -0.4t\Big|_{0}^{t} \Rightarrow -\frac{1}{2v^2} - \left[\frac{-1}{2*60^2}\right] = -0.4t - 0 \Rightarrow \frac{-1}{2v^2} + \frac{1}{7200} = \cdots$$

$$\Rightarrow \frac{1}{2v^2} = 0.4t + \frac{1}{7200}$$

the velocity after 4s $\rightarrow \frac{1}{2v^2} = 0.4*4 + \frac{1}{7200} \Rightarrow 2v^2 = \frac{1}{1.6} \Rightarrow v^2 = 0.31$

$$\Rightarrow v = \pm\sqrt{0.31} \simeq \pm 0.56$$

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(2) $\rightarrow t = 4$ بإيجاد السرعة عند

$$\boxed{\frac{1}{2v^2} = 0.4t + \frac{1}{7200}}$$

بتوحيد المقامات

$$\frac{1}{2v^2} = \frac{2880t}{7200} + \frac{1}{7200} \Rightarrow 2v^2 = \frac{7200}{2880t+1} \Rightarrow v^2 = \frac{3600}{2880t+1}$$

$$\Rightarrow v = \sqrt{\frac{3600}{2880t+1}} = \frac{60}{\sqrt{2880t+1}} = 60(2880t+1)^{-1/2}$$

نستخدم $v = \frac{ds}{dt}$ بالتعويض $\rightarrow 60(2880t+1)^{-1/2} \frac{ds}{dt}$ فصل المتغيرات $\rightarrow \int_0^s ds = \int_0^4 60(2880t+1)^{-1/2} dt$

بأخذ التكامل $\rightarrow s\Big|_0^s = \frac{60}{2880}\int_0^4 2880(2880t+1)^{-1/2} dt \Rightarrow s = \frac{1}{48}\frac{(2880t+1)^{1/2}}{1/2}\Big|_0^4$

$$\therefore s = \frac{1}{48}\frac{(2880*4+1)^{1/2}}{1/2} \simeq 4.47$$